

诚信应考,考试作弊将带来严重后果!

# 华南理工大学期末考试

## 《20241205 Calculus Review Integral》试卷

- 注意事项: 1. 考前请将密封线内填写清楚;  
2. 所有答案请直接答在试卷上(或答题纸上);  
3. 考试形式: 闭卷;  
4. 本试卷共 8 大题, 满分 100 分, 考试时间 120 分钟。

| 题号  | 1, 2 | 3, 4 | 5, 6 | 7, 8 | 9, 10 | 总分 |
|-----|------|------|------|------|-------|----|
| 得分  |      |      |      |      |       |    |
| 评卷人 |      |      |      |      |       |    |

### 1. Answer the questions (26):

(1) Find the integral  $\int \frac{e^x}{4+9e^{2x}} dx$

(2) Find the integral  $\int_1^{16} \arctan \sqrt{\sqrt{x}-1} dx$

(3) Find the integral  $\int_0^\infty e^{-3x} \sin x dx$ ;

(4)  $f(x)$  is continuous, try to prove that

$$\int_1^a f\left(x^2 + \frac{a^2}{x^2}\right) \frac{dx}{x} = \int_1^a f\left(x + \frac{a^2}{x}\right) \frac{dx}{x}.$$

(5) Find the integral  $\int \frac{\arctan x}{x^2(1+x^2)} dx$ .

(6) Find the integral  $\int_a^{2a} \frac{\sqrt{x^2-a^2}}{x^4} dx, (a > 0)$

(7) Find the integral  $\int \sqrt{e^x+1} dx$

(8) Find the integral  $\int \frac{x^2}{(1+x^2)^{\frac{3}{2}}} dx$

(9) Find the integral  $\int (\arcsin x)^2 dx$

(10) Find the integral  $\int \frac{1}{(x+1)^2 \sqrt{x^2+2x+2}} dx$

(12) If  $f(x)$  is continue and 2 second differentiable on  $[0, 2]$  and

$$\lim_{x \rightarrow 0^+} \frac{\ln(1 + \frac{f(x)}{x})}{\sin x} = 3, \quad \int_1^2 f(x) dx = 0$$

Find  $f'(0)$  and there exist  $\xi \in (0, 2)$ ,  $f'(\xi) + f''(\xi) = 0$

(13) If  $f(x)$  is continue on  $[0, 1]$ , show  $\int_0^1 [\int_{x^2}^{\sqrt{x}} f(t) dt] dx = \int_0^1 [\sqrt{x} - x^2] f(x) dx$ .

(15) Find the integral  $\int \frac{x \cos x}{\sin^3 x} dx$

(16) If  $f(x)$  is differentiable on  $[0, 1]$  and  $2 \int_0^{\frac{1}{2}} x^2 f(x) dx = 1$ ,  $f(1) = 1$

there exist  $\xi \in (0, 1)$ ,  $f(\xi) = -\frac{\xi}{2} f'(\xi)$